# MAKING SEVERAL ISOLATED FIBRE POINTS FINITE BY AN ETALE NEIGHBOURHOOD OF SEVERAL POINTS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 12018-12044 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 12046-12091, the complete author proof (44 body lines).

*Editorial note.* Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Etale neighbourhoods, isolated points in a fibre, and finiteness of a morphism over an etale base change; the assembly is retained.

*Source reference.* Original lemma algebra-lemma-etale-makes-quasi-finite-finite-variant: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12091–12091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-etale-makes-quasi-finite-finite-multiple-points: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11974–11981. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-realize-prescribed-residue-field-extension-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10138–10145. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01TM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12091–12091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01TH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12091–12091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma morphisms-lemma-residue-field-quasi-finite: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12091–12091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma schemes-lemma-points-fibre-product: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12091–12091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ be a morphism of schemes. Let $x_1, \ldots, x_n \in X$ be points having the same image $s$ in $S$. Assume that*

(1) *$f$ is locally of finite type, and*
(2) *$x_i \in X_s$ is isolated for $i = 1, \ldots, n$.*

*Then there exist*

(a) *an étale neighbourhood $(U, u) \to (S, s)$,*
(b) *for each $i$ an integer $m_i$ and open subschemes $V_{i,j} \subset X_U$, $j = 1, \ldots, m_i$*

*such that we have*

(i) *each $V_{i,j} \to U$ is a finite morphism,*
(ii) *there is a unique point $v_{i,j}$ of $V_{i,j}$ mapping to $u$ in $U$ with $\kappa(u) \subset \kappa(v_{i,j})$ finite purely inseparable,*
(iv) *if $v_{i,j} = v_{i',j'}$, then $i = i'$ and $j = j'$, and*
(iii) *the points $v_{i,j}$ map to $x_i$ in $X$ and no other points of $(X_U)_u$ map to $x_i$.*

*Proof.* This proof is a variant of the proof of Algebra, Lemma algebra-lemma-etale-makes-quasi-finite-finite-variant in the language of schemes. By Morphisms, Lemma 01TH the morphism $f$ is quasi-finite at each of the points $x_i$. Hence $\kappa(s) \subset \kappa(x_i)$ is finite for each $i$ (Morphisms, Lemma morphisms-lemma-residue-field-quasi-finite). For each $i$, let $\kappa(s) \subset L_i \subset \kappa(x_i)$ be the subfield such that $L_i/\kappa(s)$ is separable, and $\kappa(x_i)/L_i$ is purely inseparable. Choose a finite Galois extension $L/\kappa(s)$ such that there exist $\kappa(s)$-embeddings $L_i \to L$ for $i = 1, \ldots, n$. Choose an étale neighbourhood $(U, u) \to (S, s)$ such that $L \cong \kappa(u)$ as $\kappa(s)$-extensions (Lemma lemma-realize-prescribed-residue-field-extension-etale).

Let $y_{i,j}$, $j = 1, \ldots, m_i$ be the points of $X_U$ lying over $x_i \in X$ and $u \in U$. By Schemes, Lemma schemes-lemma-points-fibre-product these points $y_{i,j}$ correspond exactly to the primes in the rings $\kappa(u) \otimes_{\kappa(s)} \kappa(x_i)$. This also explains why there are finitely many; in fact $m_i = [L_i : \kappa(s)]$ but we do not need this. By our choice of $L$ (and elementary field theory) we see that $\kappa(u) \subset \kappa(y_{i,j})$ is finite purely inseparable for each pair $i, j$. Also, by Morphisms, Lemma 01TM for example, the morphism $X_U \to U$ is quasi-finite at the points $y_{i,j}$ for all $i, j$.

Apply Lemma lemma-etale-makes-quasi-finite-finite-multiple-points to the morphism $X_U \to U$, the point $u \in U$ and the points $y_{i,j} \in (X_U)_u$. This gives an étale neighbourhood $(U', u') \to (U, u)$ with $\kappa(u) = \kappa(u')$ and opens $V_{i,j} \subset X_{U'}$ with the properties (i), (ii), and (iii) of that lemma. We claim that the étale neighbourhood $(U', u') \to (S, s)$ and the opens $V_{i,j} \subset X_{U'}$ are a solution to the problem posed by the lemma. We omit the verifications. □